EEZ DIB2 specification for BB3+

Current version	*v2*
Status	*Active*
File repository	*https://github.com/eez-open/eez-bb3plus/tree/main/DIB*
License	*TAPR v1.0*
Contributions	*C4.1 (Collective Code Construction Contract)*
Revision history	*2026-10-10: initial version* *2026-10-31: BOOTn functionality updated* *2026-11-01: PCB layout for "Non-power" module added*

DIB2: An Open-Source Standard for Modular Test & Measurement

The DIB2 (*DIY Instrument Bus v2.0*) specification defines a modular, plug-and-play architecture for open-source test & measurement instruments. Unlike proprietary standards (PXI, PXIe, VXI) or legacy interfaces (GPIB), DIB2 is fully open-source, hardware-certified for CE/EMC compliance, and designed for accessibility; making professional-grade instrumentation feasible for educational labs, maker spaces, and specialized testing applications. Critically, DIB2 modules are managed and orchestrated through EEZ Studio, a free, open-source GUI development and T&M automation platform that rivals commercial solutions like LabVIEW, VEE, and MATLAB.

The DIB2 version is used in the EEZ BB3+ T&M chassis and is the successor to the existing one used in the EEZ BB3. Backward compatibility is not preserved because of the more advanced architecture used and a larger number of communication interfaces.

Key Design Principles

DIB2 combines:

- Modular flexibility: Mix-and-match modules (half-width, full-width, and 2/3-width) in a 5-slot backplane
- Modern high-speed interfaces: USB 3.x SuperSpeed and PCIe x1 for streaming data; UART/I2C/SPI for control
- Synchronized multi-instrument measurements: 10 MHz distributed clock + trigger bus for coordinated operations
- Inherent safety: Firmware-controlled relay-based power coupling eliminates external wiring hazards
- Unified software stack: EEZ Studio provides drag-and-drop GUI design, visual flow programming (EEZ Flow), and remote T&M automation—eliminating vendor lock-in
- Accessibility: Low cost connectors, open KiCad designs, OSHWA-certified hardware modules, CE-marked reference implementations

DIB2 details

The specification includes two physical connectors, dimensions and front masks of the DIB2 module.

Signal connector

The signal connector is used to communicate with the main processor in the CM5 socket as well as with the SYSMCU located on the BB3+ mainboard. On the DIB2 module, it is a PCB edge connector that plugs into a PCIe x4 64-pin connector. Examples of mainboard/backplane connector models (gold plating of 0.25 µm or thicker, not „flash") are 10018783-10101TLF and 10018783-11101TLF.

Note: Standard PCIe x4 cards must not be inserted into the DBI2 bus even though they have a compatible connector, because the signal and power pinouts are different. PCIe x1 is provided on DIB2 but with a different pinout as shown below.

The pin mapping is shown in Fig. 1.

The connector features two rows; therefore, the pinout description is divided into two columns (*A* and *B*). The direction (*Dir* column), where applicable, indicates the unidirectional signal flow relative to the inserted module, rather than the mainboard.

DIB2 offers 4 different power rails:

- +12V (4 pins)
- +5V (2 pins)
- +3.3V (2 pins)
- +Vaux (Battery/Supercap backup, max. 3.3 V)

Common GND (mainboard/backplane) is available on 11 pins. PE (Protective Earth) is available on one pin (A1) and has no direct contact with GND.

The BB3+ mainboard uses +12 V as the main power source from which all other voltages are derived, including the +5 V and +3.3 V present on DIB2. The standard total power is 45 W, it can be upgraded to 60 W.

The following interfaces are available for communication with the module:

Interface	Note
I2C/SMbus	shared among all modules, under the control of the CM5 processor.
UART	3.3V logic level serial asynchronous communication, shared between all modules and also connected to DIN1 and DOUT1 of the isolated I/O ports on the rear panel of the BB3+. *IMPORTANT: For proper operation, RX and TX must be swapped at the module side: the TX coming from the CM5 must be connected to the module's RX, and vice versa.*
SPI	Three SPI channels; slots 1 and 3 share the first channel, slots 2 and 5 share the second channel, and slot 4 has a dedicated third channel. Each DIB2 slot has two dedicated SPI Chip Select (CS) inputs; CSAn and CSBn and Interrupt Request (IRQ) output (n = 1 – 5).
USB FS/HS	Each module has a dedicated USB port, requires the implementation of a USB 2 hub on the mainboard/backplane.
USB SS	Each module has a dedicated USB port, requires the implementation of a USB 3.x hub on the mainboard/backplane. Typically, the same hub can host both USB 2.0 and USB 3.0 ports. BB3+ uses Microchip USB5807C 7-port USB3 hub featuring *FlexConnect* that enables flexible port switching and host management. It allows a USB hub to dynamically switch between two USB hosts, making any port capable of acting as a host via the FlexConnect feature. Therefore, the host (upstream) can be the CM5 processor (USB Port #0) or a PC when connected to the bottom USB socket (Port #1) on the rear panel of the BB3+.
PCIe	PCIe x1 interface with PCIe clock request and PCIe WAKE control signals. Requires the implementation of a PCIe switch or Root Complex on the mainboard/backplane. It has not been implemented in BB3+.
Ethernet (optional)	4-lane Ethernet; requires the implementation of an Ethernet switch on the mainboard/backplane. It has not been implemented in BB3+.

Signal connector							*Usage: Mandatory*
	A	Dir	Description		B	Dir	Description
1	PE	I	Protective Earth	1	+12V	I	DC power
2	+12V	I	DC power	2	+12V	I	DC power
3	+12V	I	DC power	3	GND		
4	GND			4	+3V3	I	DC power
5	+5V	I	DC power	5	+3V3	I	DC power
6	+5V	I	DC power	6	+VAUX	I	Backup DC power
7	GND			7	GND		
8	SHUTDOWN	I	Shutdown/Low-power (active low)	8	RESETn	I	Module reset (active low)
9	SMBUS_ALERT	O	Shared SMbus alert (active low)	9	NFAULT	I/O OD	Fault (open drain)
10	SCL/SMBCLK	I	Shared I²C/SMbus Clock	10	SDA/SMBDAT	I/O	Shared I²C/SMbus Data
11	GND			11	GND		
12	GND			12	GND		
13	RSVD1		Reserved / Ethernet TXRX_A-	13	OE_SYNC/TINn	I	Module Trigger In
14	RSVD2		Reserved / Ethernet TXRX_A+	14	AC_FREQ	I	AC Mains frequency
15	RSVD3		Reserved / Ethernet TXRX_B-	15	TOUTn	O	Module Trigger Out
16	RSVD4		Reserved / Ethernet TXRX_B+	16	REFCLK+	I	SE or CLK+ 10MHz reference
17	RSVD5		Reserved / Ethernet TXRX_C-	17	REFCLK-	I	CLK- 10 MHz reference
18	RSVD6		Reserved / Ethernet TXRX_C+	18	UART_RX*	O	Shared UART RX
19	RSVD7		Reserved / Ethernet TXRX_D-	19	UART_TX*	I	Shared UART TX
20	RSVD8		Reserved / Ethernet TXRX_D+	20	BOOTn	I	Module bootloader select
21	SCLK	I	Shared SPI CLK	21	GND		
22	MOSI	I	Shared MOSI	22	PCIe_CLK+	I	PCIe clock
23	MISO	O	Shared MISO	23	PCIe_CLK-	I	
24	IRQn	O	Module IRQ	24	GND		
25	CSAn	I	Module Chip select A	25	PCIe_RX-	I	PCIe RX
26	CSBn	I	Module Chip select B	26	PCIe_RX+	I	
27	HSD+	I/O	USB 2.0	27	PCIe_TX-	O	PCIe TX
28	HSD-	I/O		28	PCIe_TX+	O	
29	SSTX+	O	USB 3.0 Tx	29	PCIe_CLKREQ	I	PCIe clock request
30	SSTX-	O		30	PCIe_WAKE	I	PCIe wake
31	SSRX+	I	USB 3.0 Rx	31	GND		
32	SSRX-	I		32	ETH_SYNC	O	IEEE 1588 sync

Fig. 1: DIB2 signal connector pin mappings

The following can be used to control the DIB2 module:

Name	Function
SHUTDOWN	Active low, shared. It is used as a signal to the module to enter low-power mode, but it is not mandatory. It is controlled by the SYSMCU on the BB3+ mainboard.
RESETn	Active low, dedicated to each module (n = 1 – 5), under the control of the SYSMCU. For example, RESET1 is used to reset the module in slot #1. Upon power-up, all resets are active. It is controlled by the SYSMCU on the BB3+ mainboard.
NFAULT	An active-low, open-drain signal indicates that one or more modules are in fault mode. The BB3+ SYSMCU processes this signal.
OE_SYNC/TIN	It is used to synchronize activation of multiple power outputs (Output Enable) or to simultaneously activate other activities (start ADC, DAC on multiple modules, etc.). It is controlled by the CM5 processor.
TOUT	Trigger output, 3-state, shared among all modules, and optionally the RS485_TX_EN/DOUT2 output on the CM5. Only one module or the CM5 is permitted to generate this signal at any given time; all others must be in the Hi-Z state.
REFCLKn+	Single-ended (SE) or positive 10 MHz low-jitter buffered clock source. It is recommended that the processor and peripherals on the module use this as the master clock if better matching of measured data and activity between modules is required. *BB3+ offers as an option that the module on slot #5 generates a reference clock of high precision and stability.*
REFCLKn-	Negative buffered clock source. If used as single-ended, REFCLKn- can be on

	GND.
BOOTn	This is a bidirectional signal, and controlled by the SYSMCU on the BB3+ mainboard. At power-up it is set as an input. Then all slots are scanned and detected where DIB2 modules are present. For this reason, the module should have a pull-down resistor installed (e.g., 4.7 kΩ to 10 kΩ) for detection to be successful. When configured as output, it enables the on-board MCU to enter bootloader mode during reset for firmware uploading/flashing purposes. Communication in bootloader mode depends on the wiring of the on-board MCU and can be realized via UART, SPI or USB2. *Note: each module can be flashed separately. During flashing, it is recommended that all other modules be inactive (use RESETn and/or SHUTDOWN).*
ETH_SYNC	Directly connected to the output (pin 18) of the same name on the CM5 module. Currently not in use.

Power connector (optional)

BB3+ mainboard has power relays that enable different couplings of the DIB2 module's power output without the need to use external wiring. This reduces the possibility of errors due to incorrect wiring as well as better control of the output values when the power outputs are connected in series or in parallel.

IMPORTANT: The power module (e.g., DC power source, electronic load, SMU) must have floating outputs/inputs if series, parallel, or split-rail coupling is used.

To enable the aforementioned output coupling, the module must have its power lines connected to the DIB2 power module.

Power connectors are only available for modules in slots #1, #2 and #4.

The pin mapping is shown in Fig. 2.

Name	Function
IN+	5 pins for connecting the positive voltage from the DIB2 module to the coupling relays on the mainboard.
OUT+	5 pins for connecting the positive voltage that returns from the coupling relays on the mainboard to the output power terminal of the DIB2 module.
IN-	4 pins for connecting the negative voltage from the DIB2 module to the coupling relays on the mainboard.
OUT-	4 pins for connecting the negative voltage that returns from the coupling relays on the mainboard to the output power terminal of the DIB2 module.
SENSE+	The positive power sense input, which is connected to the rear panel terminal for slots #1 and #2.
SENSE-	The negative power sense input, which is connected to the rear panel terminal for slots #1 and #2.

Power connector (2 x 11-pin) *Usage: Optional*

	A	Dir			B	Dir	
1	IN+	I	Power positive input	2	IN+	I	Power positive input
3	IN+	I	Power positive input	4	IN+	I	Power positive input
5	IN+	I	Power positive input	6	OUT+	O	Power positive output
7	OUT+	O	Power positive output	8	OUT+	O	Power positive output
9	OUT+	O	Power positive output	10	OUT+	O	Power positive output
11	OUT-	O	Power negative output	12	OUT-	O	Power negative output
13	OUT-	O	Power negative output	14	OUT-	O	Power negative output
15	IN-	I	Power negative input	16	IN-	I	Power negative input
17	IN-	I	Power negative input	18	IN-	I	Power negative input
19	SENSE-	I	Sense negative input	20	SENSE+	I	Sense negative input

Fig. 2: DIB2 Power connector pin mappings (optional)

The power connector for the DIB2 backplane is a 20-pin, dual-row header with a 0.1" (2.54 mm) pitch, such as PH2-20-UA, 10129381-922002BLF, TSW-110-07-L-D or 77313-124-22LF.

IMPORTANT: No component taller than 5 mm (the height of the power connector) may be placed in the module PCB area where the power connector is located, even if the connector is not being used.

DIB2 module PCB dimensions

The DIB2 specifies only the physical dimensions of the peripheral module and positions of related mandatory and optional connectors (i.e. 64-pin peripheral module PCB edge connector and 20-pin power module). Dimensions of other parts of the system can differ from project to project depending of e.g. chosen enclosure dictated by builder budget and appearance preferences.

If the BB3+ backplane is used, you should pay attention to the fact that there are two PCB templates: one for power modules that also use the DIB2 power connector and the other for "non-power" modules that, if you want to install them without restrictions in any of the 5 slots, must have the bottom right part of the PCB removed in order to avoid a collision with the coupling relays installed in that part of the backplane.

The layout of the PCB for both versions is shown in Fig. 3 and Fig. 4

Thanks to the fact that DIB2 connectors are located at the bottom end, the PCB width can vary and it could be anything from 90 to 185 mm. In fact it could be even wider, but this is recommended if the EEZ BB3+ enclosure is used. Recommended PCB height is 95 mm but it can also vary in accordance with selected enclosure, and that is a max. height for the EEZ BB3+ enclosure.

DIB2 introduces a half distance (17.75 mm) in addition to the existing distance between two modules of 35.5 mm (7 HP). Now it is also possible to use up to two 3/2 (53.25 mm) width modules.

The BB3+ mainboard can now accommodate up to 5 modules in different combinations:

Module's count	Configuration
1	Half-width module in any of the 5 slots; full-width in slots #1, #2, #3 (if output coupling is not used) and #4; 3/2-width module in slots #1 or #3.
2	The same applies as above, but for two modules.
3	Three half-width modules in any of the 5 slots; three full-width modules in slots 1, 2, and 4. The following configurations are also possible: two half-width modules and one full-height module; two half-width modules and one 3/4-width module; or one half-width, one full-width, and one 3/4-width module.
4	Four half-width modules, or three half-width modules and one full-width module. Two half-width and two full-width modules (Full-width modules can be installed in all slots except #5 and #3 if output coupling is needed).
5	Five half-width modules only.

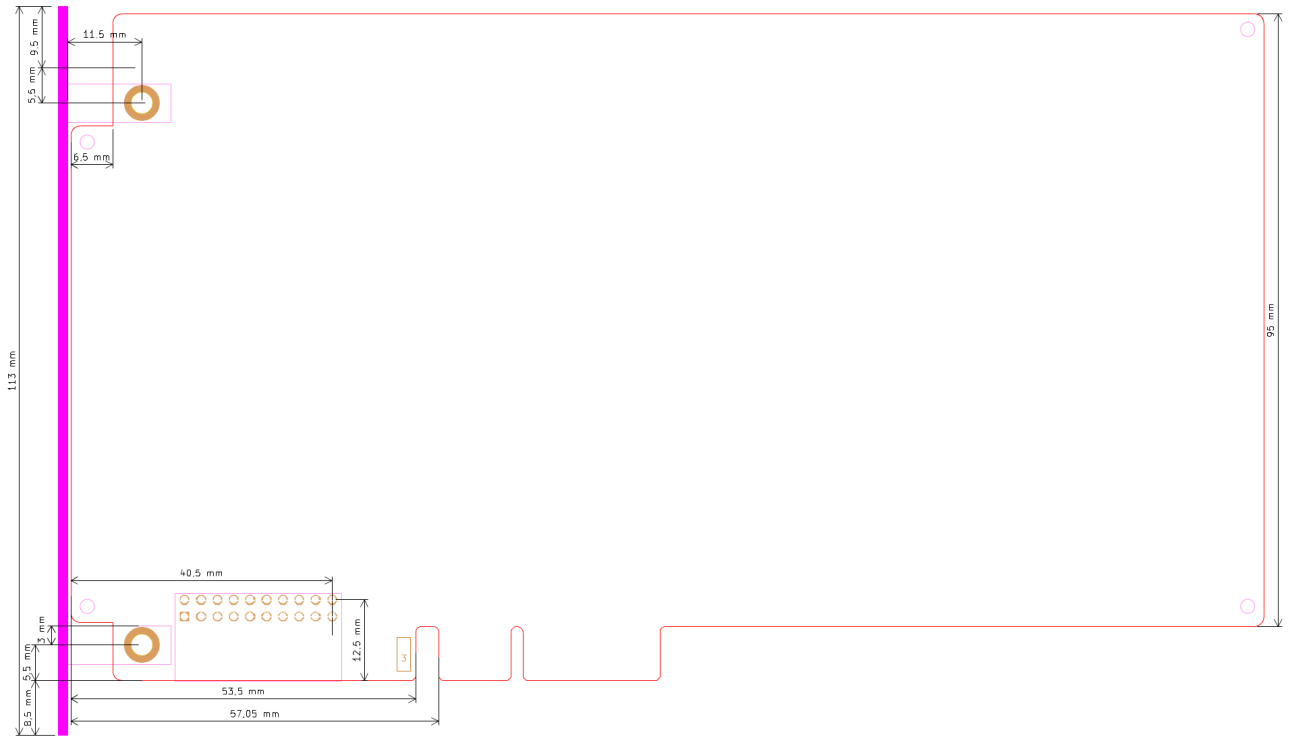


Fig. 3: DIB2 module dimensions

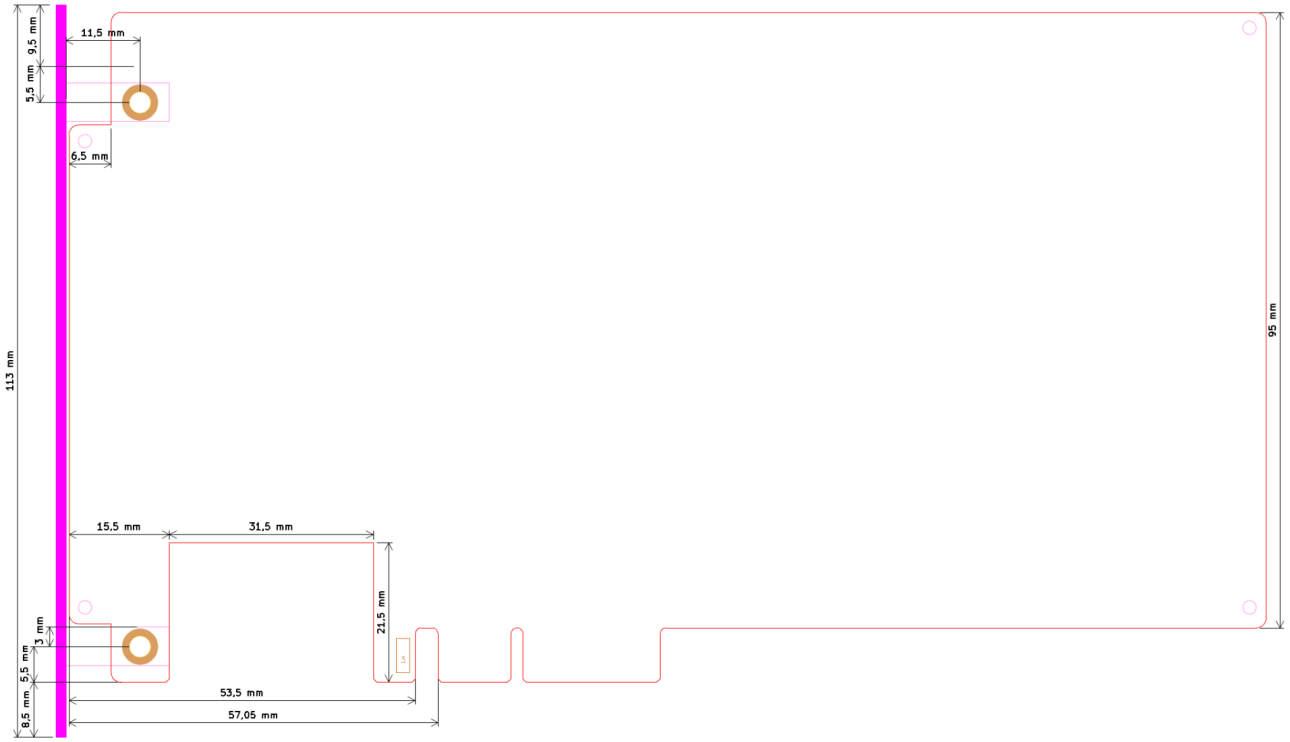


Fig. 4: DIB2 non-power module dimensions

The KiCad template for DIB2 module is available at https://github.com/eez-open/dib-kicad-template

DIB2 front panels

The BB3+ enclosure and mainboard design specifies a front panel thickness of 1.6 mm, which corresponds to the thickness of a standard PCB. This makes it possible to use PCB manufacturers to produce front panels at very affordable prices. The PCB material can be standard FR4, ideally with a blue solder mask and white silkscreen. However, since variations in the shade of blue can occur even with the same manufacturer, it is recommended to use an aluminum PCB version (with black silkscreen),

where color inconsistencies should not be an issue.

Fig. 5 and 6 show the dimensions of the half-width and full-width front masks. The half-width front mask has one mounting position, while the full-width has three positions for attachment to the BB3+ enclosure. A countersunk self-tapping screw, such as the B2.2X6.5/B14066 (Bossard), is used for mounting.

Both front masks also feature two mounting holes for attachment to the DIB2 module. M3 PH1 screws (5–8 mm in length) can be used to secure them to the PCB-MB-01 (Essentra) plastic mounting block.

KiCad template for DIB2 front masks is available at https://github.com/eez-open/eez-bb3plus/tree/main/KiCad/Front%20masks and silkscreens at https://github.com/eez-open/eez-bb3plus/tree/main/fabrication/silkscreens (.png must be imported into *KiCad PCB Editor* using *KiCad Image Converter*).

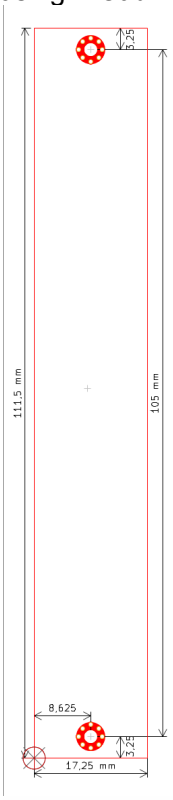


Fig. 5: Half-width DIB2 module front mask

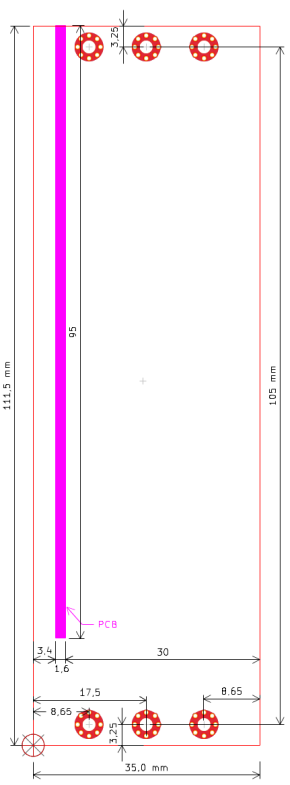


Fig. 6: Full-width DIB2 module front mask

DIB2 Strengths

Feature	Implementation	Competitive Position
Multi-Protocol Communication	I2C (shared), UART (shared), 3× SPI channels, USB 2.0/3.0 (dedicated per slot), PCIe x1	Rivals PXI/PXIe; unique among hobbyist standards
High-Speed Data	USB 3.x SuperSpeed + PCIe x1 support	Suitable for digitizers, waveform generators, data acquisition at modern speeds
Clock Distribution	10 MHz reference clock (single-ended + differential), with optional high-precision oscillator from slot #5	Enables synchronized multi-module measurements; professional-grade timing
Trigger Synchronization	OE_SYNC/TIN (input) + TOUT (output, 3-state, open-drain)	Cross-module coordination for power ramp, ADC start, etc.
Flexible Power Architecture	Mainboard relays for series/parallel/split-rail coupling; 45–60 W power budget	Eliminates external wiring errors; firmware-controlled safety
Modular Design Flexibility	Half-width (17.75 mm), full-width	Unprecedented density and

	(35.5 mm), 3/2-width (53.25 mm) modules; up to 5 slots	configuration flexibility for custom systems
Modern Isolation Strategy	Isolation implemented per-module as needed (avoids one-size-fits-all overhead)	Cost-effective for low-isolation modules; flexible for high-V designs
Firmware Management	Dedicated BOOTn per slot; can flash individual modules independently Robust OTA updates; debugging without removing modules	

Competitive Advantages vs. Proprietary Standards & Software

Metric	DIB2 + EEZ Studio	PXI + LabVIEW	PXIe + VEE	GPIB (legacy)	Custom Python
Hardware cost	$$	$$$$$	$$$$$$	$$	$$
Software cost	FREE	$$$$	$$$	FREE (old)	FREE (time)
High-speed I/O	✓ USB 3.x, PCIe x1	✗	✓✓✓	✗	✓
Modularity	✓ Half / full / 3/2-width	✓	✓	✓ (VXI racks)	~
Synchronized measurement	✓ 10 MHz clock + trigger	✓✓	✓✓✓	~	~
Open-source (hardware)	✓✓✓	✗	✗	✗	✓
Open-source (software)	✓✓✓	✗	✗	✗	✓
No-code/Low-code GUI	✓ (EEZ *Flow*)	✓	✓✓	✗	✗
Vendor lock-in	None	High	High	Obsolete	None
Learning curve (non-programmers)	Shallow	Steep	Moderate	Shallow (outdated)	Steep
CE-certified designs	✓ (reference modules)	✓	✓	✗ (pre-2000s)	✗
Educational accessibility	✓✓✓	✗	✗	✗	✓

Key insight: DIB2 + EEZ Studio is the only combination that provides:

- Hardware modularity and open-source ecosystem
- Professional T&M capabilities and zero licensing costs
- Modern high-speed I/O and legacy instrument support (via VISA/GPIB adapter)
- Drag-and-drop GUI design and industrial-grade automation scripting

Ecosystem Maturity

DIB2 is not theoretical – three production-ready modules, one certified mainboard, a complete software stack, and an active developer community demonstrate real-world applicability:

- ✓ Open KiCad schematics for all reference designs
- ✓ OSHWA hardware certification (HR000123, HR000120, HR000124)
- ✓ CE marking + EMC pre-certification reports
- ✓ Full firmware source code
- ✓ EEZ Studio with 40+ project templates and IEXT catalog
- ✓ Long-term funding via NLnet foundation
- ✓ Active GitHub repositories (50+ contributors)

Third-party developers can:

1. Fork reference designs (DCP405+ or MIO168+) as a template
2. Adapt DIB2 pinout for their specific instrument
3. Create IEXT instrument definitions in EEZ Studio
4. Validate against the open specification
5. Contribute back to the ecosystem (IEXTs, firmware, EEZ Studio plugins)

EEZ Studio Core Features for DIB2

- EEZ *Flow*: Visual, low-code/no-code flowchart programming for complex multi-instrument automation
- IEXT (*Instrument EXTensions*): Pre-built instrument definitions for BB3+, DCP405+, MIO168+; can define custom SCPI commands
- Session-oriented data logging: Capture measurements, create searchable databases, export to CSV
- Offline waveform analysis: FFT, harmonic analysis, period/frequency calculation
- Instrument Terminal: Interactive SCPI command entry with context-sensitive help
- Multi-language support: Deploy custom GUIs in any language (LTR/RTL)
- C++ code generation: Export GUIs for embedded systems (e.g., BB3+ touchscreen native apps)
- No royalties: Output code is royalty-free (MIT, BSD 2.0, or Public Domain)

Typical Applications

DIB2 + EEZ Studio is particularly suited for:

Application	Why DIB2 Excels
Educational labs	Open-source, affordable, easy to repair and modify; EEZ Studio teaches modern GUI/automation concepts
Renewable energy testing	Multiple power modules + EEZ *Flow* for multi-phase system validation; synchronized measurements via 10 MHz clock
Power electronics development	Accurate isolated measurements + oscilloscope-class waveform capture; FFT/harmonics analysis in EEZ Studio
Automated test & measurement (ATE)	Synchronized multi-module operation + visual programming = rapid deployment; unattended logging via Linux CM5
IoT sensor validation	Mixed I/O module for digital/analog stimulus; EEZ Studio dashboards for real-time trending
RF/microwave front-ends	USB 3.x streaming for high-sample-rate digitizers; custom GUIs built in hours vs. weeks in LabVIEW
Production line testing	Low-cost multi-channel measurements; EEZ *Flow* automation for pass/fail sequences; no per-seat licensing
Legacy instrument control	VISA/GPIB adapters allow DIB2 to coexist with older equipment; EEZ Studio unified interface for all devices